
Robotics & State Estimation (RiSE) Tutorials

Lecture Notes

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Robotics & State Estimation Tutorials

<https://github.com/yangyulin/rise-tutorial>



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1 Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems

1.1 Probability

1.1.1 Definition

For an event A , the probability $P(A)$ is

$$P(A) = \frac{N_A}{N}, \quad (1)$$

where N_A is the number of occurrences of A and N is the number of all possible outcomes.

1.1.2 Conditional probability

The probability of event A given that event B is true is

$$P(A | B) = \frac{P(A, B)}{P(B)}, \quad P(B) \neq 0, \quad (2)$$

which yields the product rule

$$P(A, B) = P(A | B) P(B) \quad (3)$$

$$= P(B | A) P(A). \quad (4)$$

1.1.3 Total probability theorem

If $\{A_1, \dots, A_n\}$ is a partition (mutually exclusive, $P(A_i, A_j) = 0$ for $i \neq j$) and B is an arbitrary event, then

$$P(B) = P(B | A_1)P(A_1) + \dots + P(B | A_n)P(A_n) = \sum_i P(B | A_i) P(A_i). \quad (5)$$

1.1.4 Bayes' theorem

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{P(B)} = \frac{P(B | A_i) P(A_i)}{\sum_i P(B | A_i) P(A_i)}. \quad (6)$$

1.1.5 Independence

With the inclusion–exclusion relation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, two events A and B are independent when

$$P(A, B) = P(A | B) P(B) = P(A) P(B) = P(B | A) P(A) = P(B) P(A). \quad (7)$$

1.2 Random Variables

A random variable defines a function that quantifies an experiment. For a sample space $S = \{s_1, \dots, s_n\}$, a random variable is a map $x : S \rightarrow \mathbb{R}$ with $x(s_i) = x_i$.

1.2.1 Distribution

The (cumulative) distribution of a random variable x is

$$F_x(x) = P(x \leq x), \quad (8)$$

i.e. the probability that x takes a value lower than or equal to x .

1.2.2 Probability density function (pdf)

$$p(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_x(x) - F_x(x - \varepsilon)}{\varepsilon} = \frac{dF_x(x)}{dx} = \frac{dP(x \leq x)}{dx}, \quad (9)$$

hence

$$P(x \leq x) = \int_{-\infty}^x p(\xi) d\xi, \quad P(x \leq \infty) = 1 = \int_{-\infty}^{+\infty} p(\xi) d\xi, \quad (10)$$

and for an interval

$$P(x_1 \leq x < x_2) = \int_{x_1}^{x_2} p(\xi) d\xi. \quad (11)$$

1.2.3 Example distributions

- Gaussian: $x \sim \mathcal{N}(m, \sigma) \iff p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}$.
- Uniform: $p(x) = \begin{cases} 0 & x < x_1 \\ \frac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$

1.2.4 Conditional distribution and density

$$P(x \leq x | A) = \frac{P(x \leq x, A)}{P(A)}, \quad p(x | A) = \frac{dP(x \leq x | A)}{dx}. \quad (12)$$

1.2.5 Total probability theorem for pdfs

For a mutually exclusive partition $\{A_1, \dots, A_n\}$,

$$p(x) = p(x | A_1)P(A_1) + \dots + p(x | A_n)P(A_n). \quad (13)$$

1.2.6 Continuous Bayes' theorem

Using the infinitesimal relations $P(x = x) = p(x) \delta x$ and $P(x = x | A) = p(x | A) \delta x$,

$$P(A | x = x) = \frac{P(A, x = x)}{P(x = x)} = \frac{P(x = x | A) P(A)}{P(x = x)} = \frac{p(x | A)}{p(x)} P(A), \quad (14)$$

so that $P(A | x = x) p(x) = p(x | A) P(A)$, which gives the continuous total-probability and Bayes' relations

$$P(A) = \int_{-\infty}^{+\infty} P(A | x = x) p(x) dx, \quad (15)$$

$$p(x | A) = \frac{P(A | x = x)}{P(A)} p(x) = \frac{P(A | x = x) p(x)}{\int_{-\infty}^{+\infty} P(A | x = x) p(x) dx}. \quad (16)$$

1.3 Expectation, Variance, and Random Vectors

1.3.1 Mean

The expected value (mean) of a continuous random variable, and its conditional version, are

$$m_x = \mathbb{E}(x) = \int_{-\infty}^{+\infty} x p(x) dx, \quad m_x = \mathbb{E}(x | A) = \int_{-\infty}^{+\infty} x p(x | A) dx. \quad (17)$$

1.3.2 Variance

$$\sigma_x^2 = \mathbb{E}((x - m_x)^2) = \int_{-\infty}^{+\infty} (x - m_x)^2 p(x) dx = \mathbb{E}(x^2) - (\mathbb{E}(x))^2, \quad (18)$$

with the conditional variance

$$\sigma_{x|A}^2 = \mathbb{E}\{(x - m_{x|A})^2 | A\} = \int_{-\infty}^{+\infty} (x - m_{x|A})^2 p(x | A) dx. \quad (19)$$

1.3.3 A set of random variables

The joint pdf of $x_1, \dots, x_n$ satisfies

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p(x_1, \dots, x_n) dx_1 \cdots dx_n = 1. \quad (20)$$

If x_i, x_j are independent, $p(x_i, x_j) = p(x_i) p(x_j)$. Expectation is linear: for $y = \alpha x_i + \beta x_j$,

$$\mathbb{E}(y) = \alpha \mathbb{E}(x_i) + \beta \mathbb{E}(x_j). \quad (21)$$

1.3.4 Covariance

$$\sigma_{ij}^2 = \mathbb{E}((x_i - m_i)(x_j - m_j)) = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i) \mathbb{E}(x_j). \quad (22)$$

When $\sigma_{ij} = 0$ we have $\mathbb{E}(x_i x_j) = \mathbb{E}(x_i) \mathbb{E}(x_j)$ and x_i, x_j are *uncorrelated*.

Discussion questions.

1. If x_i, x_j are independent, then $p(x_i, x_j) = p(x_i) p(x_j) \Rightarrow \mathbb{E}(x_i x_j) = \mathbb{E}(x_i) \mathbb{E}(x_j)$ — so independence implies uncorrelated.
2. If x_i, x_j are uncorrelated, are they necessarily independent?
3. If x_i, x_j are jointly Gaussian, is independence equivalent to being uncorrelated?

1.3.5 Random vector

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$, where all elements are random variables, with mean $\mathbb{E}(\mathbf{x}) \triangleq \mathbf{m}_x$ and covariance

$$\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top), \quad (23)$$

with entries

$$P_{ii} = \mathbb{E}((x_i - m_i)^2) = \int_{-\infty}^{+\infty} (x_i - m_i)^2 p(x_i) dx_i, \quad P_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)). \quad (24)$$

Since $P_{ij} = P_{ji}$, the covariance $\mathbf{P}$ is a symmetric matrix; it is also positive (semi-)definite.

1.3.6 Gaussian vector pdf

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}^{-1}(\mathbf{x} - \mathbf{m}_x)\right), \quad (25)$$

where $|\mathbf{P}|$ is the determinant of $\mathbf{P}$, $\mathbf{m}_x = \mathbb{E}(\mathbf{x})$, and $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top)$.

Homework. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$. Define

$$q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x} - \mathbf{m}_x), \quad \mathbf{y} = \mathbf{P}_{xx}^{-1/2}(\mathbf{x} - \mathbf{m}_x), \quad q = \mathbf{y}^\top \mathbf{y} = \sum_i y_i^2.$$

What are the mean and covariance of q ?

1.4 Maximum Likelihood and Maximum a Posteriori Estimation

For a measurement model with additive noise $z_i = h_i(\mathbf{x}) + n_i$, stacked as $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, two estimators are

- **Maximum likelihood estimation (MLE):** $\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x})$.
- **Maximum a posteriori (MAP):** $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$, with $p(\mathbf{x} | \mathbf{z}) = \frac{p(\mathbf{z} | \mathbf{x}) p(\mathbf{x})}{p(\mathbf{z})}$.

For zero-mean Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{P})$,

$$\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x}) = \max_{\mathbf{x}} \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-(\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x}))\right), \quad (26)$$

which is equivalent to the weighted least-squares problem

$$\min_{\mathbf{x}} C = (\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x})). \quad (27)$$

Linearizing about $\hat{\mathbf{x}}$ with $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$, and defining the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$ and error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$,

$$\min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}})^\top \mathbf{P}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}}^\top \tilde{\mathbf{z}} - 2\tilde{\mathbf{z}}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top \mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}}). \quad (28)$$

Setting the gradient to zero gives the normal equations $\mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} = \mathbf{H}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}}$, hence

$$\boxed{\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}} = \hat{\mathbf{x}} + (\mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H})^{-1} \mathbf{H}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}}.} \quad (29)$$

This is the classical weighted least-squares solution.

Questions.

1. What is the covariance of the new estimate $\mathbf{x}^\oplus$?
2. For a prior $\mathbf{x}_p = \mathbf{x} + \mathbf{n}_p$ together with $z_i = h_i(\mathbf{x}) + n_i$, $i = 1, \dots, n$, what is the MAP solution $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$?

1.4.1 Optimization

Nonlinear least-squares problems are solved iteratively, e.g. Gauss–Newton (GN) and Levenberg–Marquardt (LM).

1.5 Linear Systems

A linear (continuous-time) state-space system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}, \quad (30)$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ the input, and $\mathbf{y}$ the output. Its discrete-time counterpart is

$$\mathbf{x}_{k+1} = \mathbf{A}'\mathbf{x}_k + \mathbf{B}'\mathbf{u}_k, \quad \mathbf{y}_{k+1} = \mathbf{C}'\mathbf{x}_{k+1} + \mathbf{D}'\mathbf{u}_{k+1}. \quad (31)$$

If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are constant, the system is *linear time-invariant* (LTI); if they depend on time, it is *linear time-variant* (LTV).

1.5.1 Homogeneous solution

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\mathbf{y} = \mathbf{C}\mathbf{x}$. From $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{0}$,

$$\frac{d}{dt}(e^{-\mathbf{A}t}\mathbf{x}) = e^{-\mathbf{A}t}\dot{\mathbf{x}} + e^{-\mathbf{A}t}(-\mathbf{A})\mathbf{x} = e^{-\mathbf{A}t}(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}) = \mathbf{0}. \quad (32)$$

Integrating from 0 to t gives $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(t_0) = \mathbf{0}$, so

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 = \Phi(t, 0)\mathbf{x}_0, \quad \Phi(t, 0) = e^{\mathbf{A}t}, \quad (33)$$

where $\Phi(t, 0)$ is the *state transition matrix*, and $\mathbf{y} = \mathbf{C}\Phi(t, 0)\mathbf{x}_0$.

1.5.2 Properties of the state transition matrix

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0), \quad \frac{d}{dt}\Phi(t, 0) = \mathbf{A}\Phi(t, 0). \quad (34)$$

The second follows from $\dot{\mathbf{x}} = \dot{\Phi}(t, 0)\mathbf{x}_0 = \mathbf{A}\mathbf{x} = \mathbf{A}\Phi(t, 0)\mathbf{x}_0$.

1.5.3 Output sequence and observability

Sampling the homogeneous response at $t_0, t_1, \dots, t_n$ with $\mathbf{x}_k = \Phi(k, 0)\mathbf{x}_0$ and $\mathbf{y}_k = \mathbf{C}\mathbf{x}_k$ stacks into

$$\begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_n \end{bmatrix} = \begin{bmatrix} \mathbf{C}\Phi(1, 0) \\ \vdots \\ \mathbf{C}\Phi(n, 0) \end{bmatrix} \mathbf{x}_0, \quad (35)$$

which relates the measured outputs to the initial state (an observability-type relation).

1.5.4 Full LTV solution

For the complete system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}(t)$, write $\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x} = \mathbf{B}(t)\mathbf{u}(t)$ and use the integrating factor $e^{-\int_0^t \mathbf{A}(\tau) d\tau}$:

$$\frac{d}{dt} \left(e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x} \right) = e^{-\int_0^t \mathbf{A}(\tau) d\tau} (\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x}) = e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{B}(t)\mathbf{u}(t). \quad (36)$$

Integrating,

$$e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\int_0^\tau \mathbf{A}(s) ds} \mathbf{B}(\tau)\mathbf{u}(\tau) d\tau, \quad (37)$$

so that, with $\Phi(t, 0) = e^{\int_0^t \mathbf{A}(\tau) d\tau}$ and $\Phi(t, \tau) = e^{\int_\tau^t \mathbf{A}(s) ds}$,

$$\mathbf{x}(t) = \Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau, \quad (38)$$

$$\mathbf{y}(t) = \mathbf{C}(t) \left(\Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau \right) + \mathbf{D}(t)\mathbf{u}(t). \quad (39)$$

This is one of the most important properties of linear systems.

1.6 Homework Problems and Solutions

1.6.1 Variance identity

Problem. Show that the variance of a scalar random variable can be written as $\sigma_x^2 = \mathbb{E}(x^2) - (\mathbb{E}(x))^2$, and discuss when it vanishes.

Solution. Expanding the definition with $m_x = \mathbb{E}(x)$,

$$\sigma_x^2 = \mathbb{E}((x - m_x)^2) = \mathbb{E}(x^2 - 2m_x x + m_x^2) = \mathbb{E}(x^2) - 2m_x \mathbb{E}(x) + m_x^2 = \mathbb{E}(x^2) - m_x^2. \quad (40)$$

Since $\sigma_x^2 \geq 0$, we have $\mathbb{E}(x^2) \geq (\mathbb{E}(x))^2$, with equality if and only if $\sigma_x^2 = 0$, i.e. $x = m_x$ almost surely (a deterministic variable) — a degenerate special case.

1.6.2 Covariance matrix identity

Problem. For a random vector $\mathbf{x}$ with mean $\mathbf{m}_x$, show that $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top)$.

Solution. Expanding the outer product and using linearity of expectation, together with $\mathbb{E}(\mathbf{x}) = \mathbf{m}_x$,

$$\mathbf{P} = \mathbb{E}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{x}^\top + \mathbf{m}_x\mathbf{m}_x^\top) \quad (41)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbf{m}_x^\top - \mathbf{m}_x\mathbb{E}(\mathbf{x}^\top) + \mathbf{m}_x\mathbf{m}_x^\top \quad (42)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{m}_x^\top + \mathbf{m}_x\mathbf{m}_x^\top = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top \quad (43)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top). \quad (44)$$

1.6.3 Scalar covariance identity

Problem. For two random variables x_i, x_j , show that $\sigma_{ij} = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j)$.

Solution. With $m_i = \mathbb{E}(x_i)$, $m_j = \mathbb{E}(x_j)$, expand

$$\sigma_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)) = \iint (x_i - m_i)(x_j - m_j) p(x_i, x_j) dx_i dx_j \quad (45)$$

$$\begin{aligned} &= \iint x_i x_j p(x_i, x_j) dx_i dx_j - m_j \iint x_i p(x_i, x_j) dx_i dx_j \\ &\quad - m_i \iint x_j p(x_i, x_j) dx_i dx_j + m_i m_j \iint p(x_i, x_j) dx_i dx_j. \end{aligned} \quad (46)$$

Using the marginalization $\int p(x_i, x_j) dx_i = p(x_j)$ (and symmetrically), together with $\iint p(x_i, x_j) dx_i dx_j = 1$,

$$m_i \iint x_j p(x_i, x_j) dx_i dx_j = m_i \int x_j p(x_j) dx_j = m_i m_j, \quad (47)$$

and likewise for the other term, so

$$\sigma_{ij} = \mathbb{E}(x_i x_j) - m_i m_j - m_i m_j + m_i m_j = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j). \quad (48)$$

1.6.4 Independence versus uncorrelatedness

Problem. Resolve the three discussion questions: (1) does independence imply uncorrelatedness? (2) does uncorrelatedness imply independence? (3) for jointly Gaussian variables, are the two equivalent?

Solution. (1) *Independent $\Rightarrow$ uncorrelated.* If $p(x_i, x_j) = p(x_i)p(x_j)$, then

$$\mathbb{E}(x_i x_j) = \iint x_i x_j p(x_i, x_j) dx_i dx_j = \int x_i p(x_i) dx_i \cdot \int x_j p(x_j) dx_j = \mathbb{E}(x_i)\mathbb{E}(x_j), \quad (49)$$

so $\sigma_{ij} = 0$.

(2) *Uncorrelated $\not\Rightarrow$ independent.* Uncorrelatedness gives only

$$\mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j) = \iint x_i x_j (p(x_i, x_j) - p(x_i)p(x_j)) dx_i dx_j = 0, \quad (50)$$

which does *not* force $p(x_i, x_j) = p(x_i)p(x_j)$ in general. Hence uncorrelated does not imply independent.

(3) *Jointly Gaussian.* For $x_i \sim \mathcal{N}(m_i, \sigma_i^2)$ and $x_j \sim \mathcal{N}(m_j, \sigma_j^2)$, the product of marginals is

$$p(x_i)p(x_j) = \frac{1}{2\pi\sigma_i\sigma_j} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (51)$$

The joint Gaussian $[x_i, x_j]^\top \sim \mathcal{N}([m_i, m_j]^\top, \mathbf{P}_{xx})$ with $\mathbf{P}_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$ is

$$p(x_i, x_j) = \frac{1}{2\pi\sqrt{|\mathbf{P}_{xx}|}} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \mathbf{P}_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (52)$$

If $\sigma_{ij} = \sigma_{ji} = 0$, then $\mathbf{P}_{xx}$ is diagonal and $p(x_i, x_j) = p(x_i)p(x_j)$. Hence, for (jointly) Gaussian variables, *independent $\Leftrightarrow$ uncorrelated*.

1.6.5 The χ^2 distribution

Problem. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$ and define $q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1} (\mathbf{x} - \mathbf{m}_x)$. Find the mean and variance of q .

Solution. Factor $\mathbf{P}_{xx} = \mathbf{L}\mathbf{L}^\top$ (Cholesky), so $\mathbf{P}_{xx}^{-1} = \mathbf{L}^{-\top} \mathbf{L}^{-1}$, and define the whitened variable $\mathbf{y} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Then

$$q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{L}^{-\top} \mathbf{L}^{-1} (\mathbf{x} - \mathbf{m}_x) = \mathbf{y}^\top \mathbf{y} = \sum_{i=1}^n y_i^2. \quad (53)$$

The whitened variable has zero mean and identity covariance,

$$\mathbf{m}_y = \mathbb{E}(\mathbf{y}) = \mathbf{L}^{-1}(\mathbb{E}(\mathbf{x}) - \mathbf{m}_x) = \mathbf{0}, \quad (54)$$

$$\mathbf{P}_{yy} = \mathbb{E}(\mathbf{y}\mathbf{y}^\top) = \mathbf{L}^{-1} \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) \mathbf{L}^{-\top} = \mathbf{L}^{-1} \mathbf{P}_{xx} \mathbf{L}^{-\top} = \mathbf{L}^{-1} \mathbf{L}\mathbf{L}^\top \mathbf{L}^{-\top} = \mathbf{I}_n, \quad (55)$$

so $y_i \sim \mathcal{N}(0, 1)$ are independent. Therefore the mean of q is

$$\mathbb{E}(q) = \mathbb{E}(\mathbf{y}^\top \mathbf{y}) = \sum_{i=1}^n \mathbb{E}(y_i^2) = n. \quad (56)$$

For the variance, with $\mathbb{E}(y_i^2) = 1$ the cross terms vanish by independence, giving

$$\sigma_{qq}^2 = \mathbb{E}((q - n)^2) = \mathbb{E}\left(\left(\sum_i (y_i^2 - 1)\right)^2\right) = \sum_i \mathbb{E}((y_i^2 - 1)^2) = \sum_i (\mathbb{E}(y_i^4) - 2\mathbb{E}(y_i^2) + 1). \quad (57)$$

The fourth moment of a standard normal follows from integration by parts (with $u = y_i^3$, $dv = y_i p(y_i) dy_i$, $v = -p(y_i)$):

$$\mathbb{E}(y_i^4) = \int_{-\infty}^{+\infty} y_i^4 p(y_i) dy_i = \underbrace{\left[-y_i^3 p(y_i)\right]_{-\infty}^{+\infty}}_{=0} + \int_{-\infty}^{+\infty} 3y_i^2 p(y_i) dy_i = 3\mathbb{E}(y_i^2) = 3. \quad (58)$$

Hence each term equals $3 - 2 + 1 = 2$, and

$$\sigma_{qq}^2 = \sum_{i=1}^n 2 = 2n. \quad (59)$$

So $q \sim \chi^2(n)$ has mean n and variance $2n$.